Simulating Loan Thresholds

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2026-06-29

A simplified situation

Imagine a machine learning algorithm that takes information about loan applicants and assigns them a credit score from 0 to 100. The credit scores are not perfect, however. Applicants with higher credit scores are more likely to repay their loan, but some people with lower credit scores will repay loans, and some with higher scores will not.

Your task is to devise a system that uses these credit scores to decide who gets a loan.

Exercise 1. How will you measure how "good" your system is? What sorts of things could you measure/quantify as an indication of how well your system works? We'll call those things metrics. Come up with at least 3 metrics.

Exercise 2. Which of your metrics do you like better? Why?

Now let's experiment a bit with some simulated data. Go to the "Attacking discrimination with smarter machine learning" page page at

https://web.archive.org/web/20260101103806/http://research.google.com/bigpicture/attacking-discrimination-in-ml/ This website is based on issues raised in M. Hardt, E. Price, and N. Srebro [1].

Loan Applicants: two scenarios

Exercise 3. Read the opening section of the web page to make sure you understand the two scenarios presented and how they are represented graphically.

Simulating Loan thresholds

Now go to the section entitled "Simulating loan thresholds." The graphs and figures in this section provide metrics, and you can slide the credit score threshold and see how these metrics change.

Exercise 4. What metrics are included on this page? How many of them are the same (or equivalent to)[1] the ones you came up with?

Exercise 5. What credit score threshold maximizes the number of correct decisions? Does that same threshold also minimize the number of incorrect decisions?

Exercise 6. What threshold maximizes profit?

Note: The app assumes a marginal gain of $300 for each loan repaid and a loss of $700 for each loan default. This is mentioned at the end of the section, but you can also figure it out from the interactive graphic. How?

Exercise 7. Are your answers to Exercise 5 and Exercise 6 the same? Why or why not?

Exercise 8. The app doesn't let you experiment with this, but suppose repaid loans had a marginal benefit of $400 and defaulted loans a marginal cost of only $600.

a. Would that lead a profit-maximizer to increase or decrease the threshold? Why?

b. How would the threshold change if you are maximizing correct decisions.

Different groups

⚠ Don't click the red buttons until directed.

Instead, just drag the threshold sliders.

Exercise 9. The text says: "In this case, the distributions of the two groups are slightly different, even though blue and orange people are equally likely to pay off a loan."

a. In what way are the distributions (of credit scores) different for blue people and orange people? (Remember your statistics? You might think about things like center, spread, shape, etc. But remember that we are comparing distributions of two variables: credit score and whether or not they repay the loan. Furthermore, the dot plots are stacking dots – light above dark – so be sure to interpret the shapes accordingly.)

b. How might you improve the graphical display to make it easier to see how the two groups differ?

c. Does the credit score do a better job for one group than for the other? What do you mean by better?

Exercise 10. Adjust the thresholds to maximize the profit that the bank gets for each group individually.

(a) What are the resulting thresholds for the two groups? (After discovering this by sliding the thresholds, you can check your answer by clicking on the red "Max Profit" button.)

[1] For our purposes, two metrics are equivalent if they rank any two things the same – that is, they always agree about which option is the better one.

(b) At those thresholds, which group is offered more loans?
(c) Of the loans that were offered, which group repays more of those loans?
(d) What is the resulting profit for the bank?
(e) Is that fair? (explain.)

Exercise 11. Adjust the thresholds to maximize profit under the constraint that blue and orange must have the same threshold. (This is tricky in the UI; hint: it must be between the two thresholds you found in the previous exercise.) Answer the same five questions (a-e) as in the previous exercise. Then check your answer by clicking the “Group Unaware” button.

Exercise 12. Starting at the “Group Unaware” thresholds, change the threshold on the orange group to increase its Positive Rate to match the Positive Rate of the blue group.

(a) What happens to the True Positive Rate?
(b) What happens to the bank’s profit?
(c) Is that good? Explain.

Exercise 13. Explore the “Demographic Parity” and “Equal Opportunity” objectives by clicking their buttons. The authors of the interactive visualization seem to have a favorite among these objectives. What reasons might someone have to agree or disagree with their assessment?

Thinking about your thinking

There are a lot of value-laden words in the questions above: words like good, better, fair, etc. Compare two ways of awarding loans (from the ones above or from other ways you come up with), one of which you think is “better” than the other. This will be more interesting if you choose two that each of strong points rather than one that you think is great and one that you think is terrible – make the decision difficult.

Exercise 14. What makes one better than the other? Imagine you had to defend the choice, what would be the main components of your argument?

Exercise 15. Are there general principles that would apply to other situations that have nothing to do with loans and credit scores?

Exercise 16. If you could wish for any feature(s) to be added to this simulation, what would it/they be?

The article

This simple simulation is a by-product of a paper: M. Hardt, E. Price, and N. Srebro [1].

Here is the abstract:

> We propose a criterion for discrimination against a specified sensitive attribute in supervised learning, where the goal is to predict some target based on available features. Assuming data about the predictor, target, and membership in the protected group are available, we show how to optimally adjust any learned predictor so as to remove discrimination according to our definition. Our framework also improves incentives by shifting the cost of poor classification from disadvantaged groups to the decision maker, who can respond by improving the classification accuracy. We enourage readers to consult the more complete manuscript on the arXiv.

Exercise 17. For each of the bolded words/phrases in the abstract, say what they correspond to in the simulation we just looked at: (a) specified sensitive attribute; (b) some target; (c) available features; (d) predictor; (e) membership in the protected group; (f) adjust (any) learned predictor.

Bibliography

[1] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in Proceedings of the 30th International Conference on Neural Information Processing Systems, in NIPS'16. Barcelona, Spain: Curran Associates Inc., 2016, pp. 3323–3331. [Online]. Available: https://dl.acm.org/doi/pdf/10.5555/3157382.3157469